Alternative Nash Solvers for Tiny Hanamikoji

CS6208 Homework 2, Task 4 (Direction D) | 9 November 2026

1. Question and equilibrium certificate

I compare counterfactual regret minimization (CFR), fictitious play (FP), and double oracle (DO) on the same single-round game. The question is how their convergence paths and computational work differ. All main runs now complete 20,000 iterations, retaining intermediate measurements; a sequence-form linear program (LP) supplies an independent numerical equilibrium reference and the submitted Task 3 policies.

The default game has neutral favors, a random starting player, 2,757 information sets and 3,672 sequences per player. Payoffs are +1 for a win, -1 for a loss, and 0 for an unresolved round. Let x,y be realization plans and A the chance-weighted payoff matrix for P1. Each plan conserves own-action reach at every information set.

SPR(x,y) = $\max_{x'} x'^{T}Ay - \min_{y'} x^{T}Ay'$. This full saddle-point residual is the sum of the two unilateral improvement opportunities, without division by two. SPR ≤ 0.002 bounds each player's possible expected-payoff gain by 0.002; it is not a win-rate error. Self-play payoff close to the game value alone does not establish equilibrium.

2. Methods, update conventions, and convergence

CFR [1]. Counterfactual values weight chance and opponent reach, without multiplying by own prefix reach. I accumulate local regret q(I,a)-v(I), then normalize positive regrets, using uniform play when all are nonpositive. Both players update simultaneously. The returned strategy averages pre-update realization plans uniformly. The average has an $O(1/\sqrt{T})$ worst-case SPR bound; finite-time exact equilibrium is not guaranteed. CFR+, alternation, and linear averaging are disabled here.

Fictitious play [2]. Both players best-respond to the opponent's previous empirical mixture and update simultaneously. A uniform prior has weight one, so the average after t updates contains t+1 observations. Realization-plan mixtures preserve own reach and implement normal-form FP without enumerating pure plans. Its empirical gap tends to zero in finite two-player zero-sum games. The two update best responses also certify SPR before the next update.

Double oracle [3]. Pools start with one mixed uniform plan and expand with pure global best responses. Two LPs solve each changed restricted game; mixtures are formed in realization space. If neither oracle finds a new plan, the profile is a full-game equilibrium under exact solves. Otherwise a new pure plan is added; finiteness guarantees eventual certification. In the full-budget experiment, real oracle rounds continue after certification, reusing unchanged restricted equilibria without redundant LPs.

LP [4]. With flow constraints Ex=e and Fy=f, dualizing the opponent's response gives P1: maximize $f^{T}q$ subject to Ex=e, $x \geq 0$, $F^{T}q \leq A^{T}x$. P2's symmetric LP minimizes $e^{T}p$ with Fy=f, $y \geq 0$ and $E^{T}p \geq Ay$. The dual vectors are unrestricted. Sparse HiGHS solves both with 10^{-9} feasibility tolerances.

3. Experimental protocol and intermediate evidence

Full-budget comparison. Each of the three methods completes 20,000 outer iterations in four scenarios, with three fresh timing repetitions: 36 runs in total. Early stopping is disabled. The 0.002 target is retained as a retrospective first observed crossing, separately from final SPR and total runtime. Equal iteration counts measure equal outer-iteration budgets, not equal amounts of computation.

The scenarios are the default neutral game, neutral games with P1 or P2 fixed as starter, and P1 starting with the G3 favor. Every method uses exact weighted chance enumeration, the same payoff scale and deterministic tie handling. Uniform initial profiles are shared, although CFR starts with zero regrets, FP with one prior observation, and DO with one candidate per player.

Before timing each scenario I discard a two-iteration warm-up of every iterative method and one full LP solve. Three repetitions rotate method order. Wall time includes construction, updates, averaging, logging, and exact certificates; it excludes tree extraction, warm-up, file export, and subsequent independent validation. Repetitions measure timing noise, not sampling uncertainty. The reference LP is timed separately without an outer-iteration budget.

CFR records SPR every 25 iterations plus entry/exit. FP and DO save the certificate from every oracle iteration. Full measured histories are stored as compressed JSON Lines in task4/results/histories/; experiments.json retains the common 25-step grid, the first crossing, and endpoint. Explicit checkpoints include 2,000, 5,000, 10,000, and 20,000. No intermediate curve is reconstructed from final values.

CFR's first observed crossing may miss an earlier transient dip between checks. FP/DO crossings use their per-iteration certificates. Auxiliary analyses are labelled separately: DO stops at numerical SPR $\leq 10^{-9}$, and LP-versus-CFR comparisons stop CFR at 0.002. They do not replace the full-budget results.

4. How correctness is verified

Check	Independent evidence
Posterior (6 tests)	Analytic hidden-card probabilities, full joint histories, unreachable events and state restoration
Best response / payoff	Exhaustive pure plans and independent response LPs; live game-tree payoff traversal
CFR, FP and DO	Known matrix equilibria, realization flows, continuation/reset and changed payoffs
Full-budget execution	Real oracle calls after DO stabilizes; actual iteration totals, complete histories and saved checkpoints
Final strategies / LP	Reloaded probabilities and SPR; primal/dual residuals and gap

All 44 tests pass. Every benchmark output is checked independently for flow error below 10^{-8} and SPR agreement within 10^{-10}. Full-budget runs must report exactly 20,000 iterations. The protected treeplex and all 80 shared starter files remain unchanged. Source hashes and the environment are recorded with the results.

5. Default-game results: full budget and first crossing

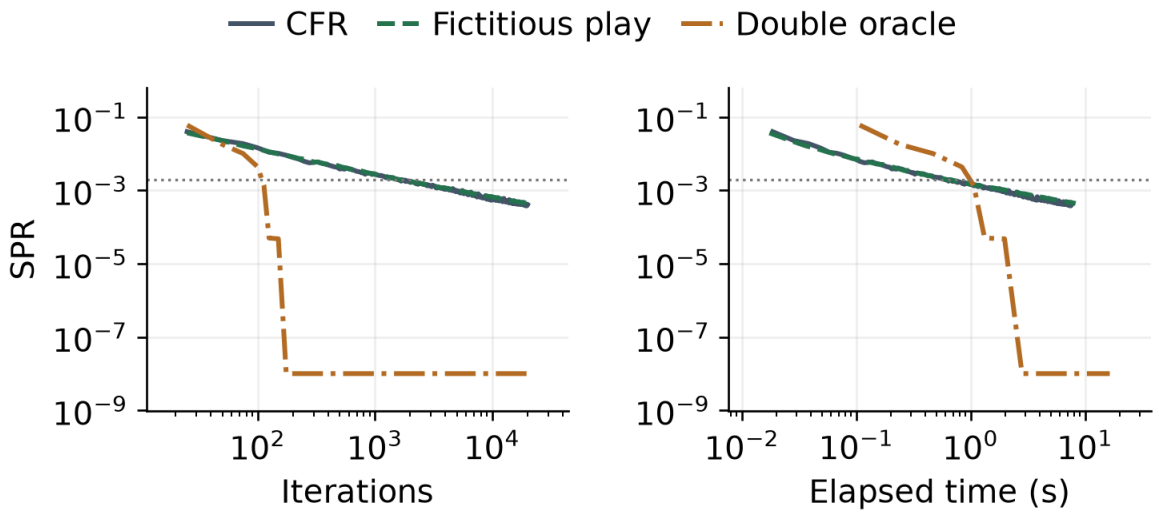


Figure 1. Recorded full-budget SPR versus iterations and constructor-plus-solve wall time, using each method's median-total-time repetition. Values below 10^{-8} are drawn at that floor for readability; raw certificates are preserved. The dotted line is the 0.002 target.

Method	SPR at 2,000	SPR at 20,000	Full time (s)	First target iteration	Time there (s)
CFR	1.7×10^{-3}	4.2×10^{-4}	7.596	1,550	0.600
FP	1.8×10^{-3}	4.4×10^{-4}	8.261	1,576	0.659
DO	4.5×10^{-15}	4.5×10^{-15}	16.413	113	1.043
LP	-	8.7×10^{-18}	0.088	-	-

Table 1. Endpoint accuracy, median full runtime, and first recorded SPR ≤ 0.002 with its median timestamp. Crossings are historical observations from uninterrupted runs. LP is a separate reference with no outer-iteration budget.

From 2,000 to 20,000 iterations, CFR's final SPR decreases by a factor of 4.01; FP's decreases by 4.02. Neither endpoint is numerically exact. DO already meets the 10^{-9} certificate in its separate precision run at iteration 122; the full run performs no further restricted LPs after iteration 258, repeatedly certifying its unchanged mixture. Its 20,000-step time is therefore not the cost required to obtain that equilibrium.

The full runs perform 40,000 CFR regret sweeps, 40,000 FP training best responses, and 40,000 DO training best responses plus 516 individual restricted LPs. DO does not solve two fresh LPs on every late iteration. The finite-range fitted SPR slopes from iteration 100 are -0.65 for CFR and -0.62 for FP; these observations are not asymptotic-rate guarantees.

6. Starting conditions, structure, and oracle cost

Scenario	CFR	FP	DO
Random start	1,550 / 0.600	1,576 / 0.659	113 / 1.043
P1 starts	1,550 / 0.361	1,576 / 0.387	59 / 0.308
P2 starts	1,550 / 0.358	1,576 / 0.378	59 / 0.295
P1 holds G3	675 / 0.158	1,386 / 0.313	66 / 0.345

Table 2. First observed target iteration / median seconds, extracted from full histories. Every method reaches the target in every scenario; the final iteration remains 20,000 in all cases.

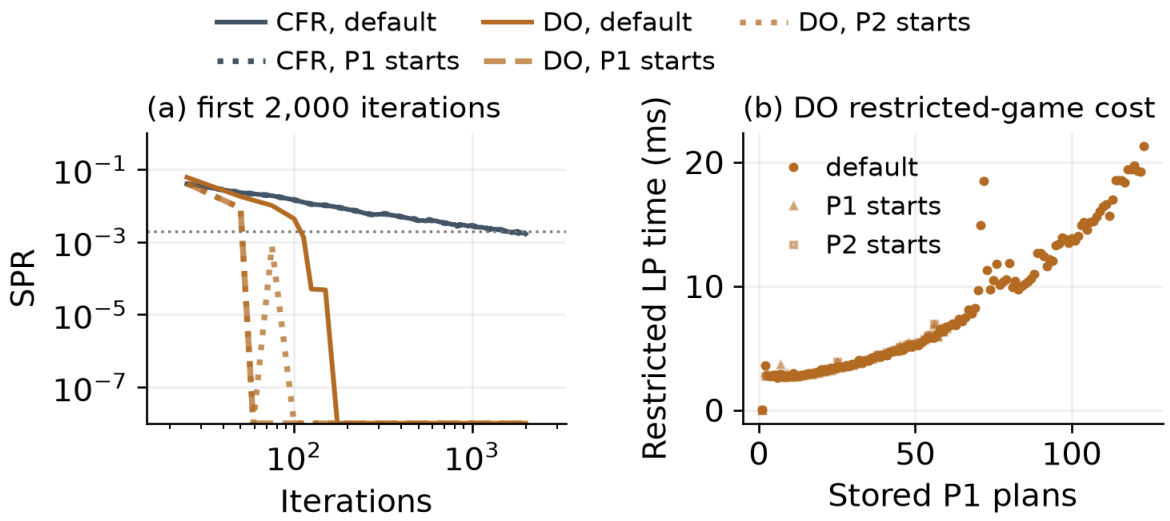


Figure 2. (a) Early certificates on the default game and its two fixed-starter halves; values below 10^{-8} share the lower edge. (b) Cost of a restricted solve, comprising two LPs, in separate DO runs to SPR $\leq 10^{-9}$.

The random-starter game is a 50/50 mixture of disjoint, observed starting-player subgames. Their LP values are +2/105 and -2/105, giving default value zero. At the target, CFR needs 1,550 iterations on the default game and 1,550/1,550 on its halves; FP needs 1,576 versus 1,576/1,576. Their local updates exploit the disjoint subtrees naturally.

DO needs 113 iterations to the target on the default game versus 59/59 on the halves. Its candidates are whole-game plans: the restricted mixtures couple decisions in both halves until enough candidates are added. This helps explain why oracle iteration counts change even when the component games remain simple. It is an empirical structural diagnosis, not a general scaling theorem.

The default precision run spends 87.1% of elapsed time in restricted LP solves and stores 123/123 candidate plans, with 14/14 weights above 10^{-12}. Stored candidates and positive-weight support are different quantities. Full-budget DO can continue discovering tied plans after first becoming numerically unexploitable. Its path depends on oracle tie-breaking and the restricted equilibrium returned by the numerical LP solver.

7. LP comparison, submitted policies, and limitations

In the separate target-accuracy comparison with P1 starting, the LP value is 0.019048, while CFR earns 0.019402. Their difference, 3.5×10^{-4}, lies below CFR's SPR of 2.0×10^{-3}. Their action probabilities need not coincide: an LP over P1's optimal face lets the probability of reserving G3, when P1 holds G1+G3, range from 0.000 to 0.417. This explicitly demonstrates equilibrium nonuniqueness.

For the default game, the LP's maximum flow residual is 3.3×10^{-16} and independent SPR is 8.7×10^{-18}. The submitted NPZ files still contain the LP policies, saved and reloaded through the supplied API; their reloaded SPR is 8.7×10^{-18}. Increasing the experimental budget does not change their solver provenance. The certificate is numerical, not symbolic exact arithmetic.

The main comparison answers accuracy after a fixed number of outer updates. Historical first-target timestamps answer a different question: effort to reach useful precision. In particular, post-convergence DO certification is deliberate experimental work, not work needed for deployment. A finite update budget and a common target do not make per-iteration work equal; both accuracy and elapsed time must be reported.

All methods still use a fully enumerated Tiny game. Larger games would stress tree memory, LP factorization, or DO candidate growth; these experiments do not establish which resource fails first. No Medium/Full or multi-round claim is made. A neutral first-round policy table requires fallback play elsewhere, without an equilibrium guarantee. Three timing repetitions are insufficient for strong claims about small speed differences.

Measurements use Apple M4, 16 GiB RAM, macOS-15.6-arm64-arm-64bit, Python 3.12.14, NumPy 2.5.4, and SciPy 1.18.1. Peak process RSS is 230.2 MiB, recorded during the main benchmark, including its histories; it is not a per-solver memory comparison. The attempt README records the reproduction sequence. Source code, compressed intermediate histories, checkpoint summaries, tests, and figures accompany this report; the earlier 2,000-iteration study is archived separately.

References

[1] M. Zinkevich, M. Johanson, M. Bowling, and C. Piccione. *Regret Minimization in Games with Incomplete Information.* NeurIPS 20, 2007. Proceedings.

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[4] B. von Stengel. *Efficient Computation of Behavior Strategies.* Games and Economic Behavior 14(2), 220-246, 1996. doi:10.1006/game.1996.0050.